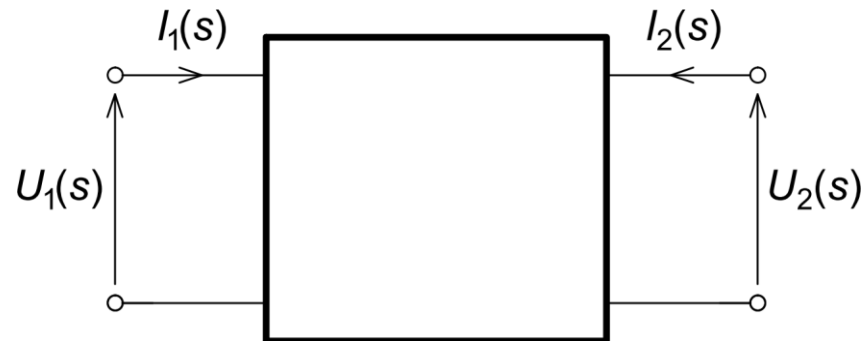


Two-Port Networks

Two-Port Networks

OPERATOR CHARACTERIZATION



Initial conditions are assumed to be zero

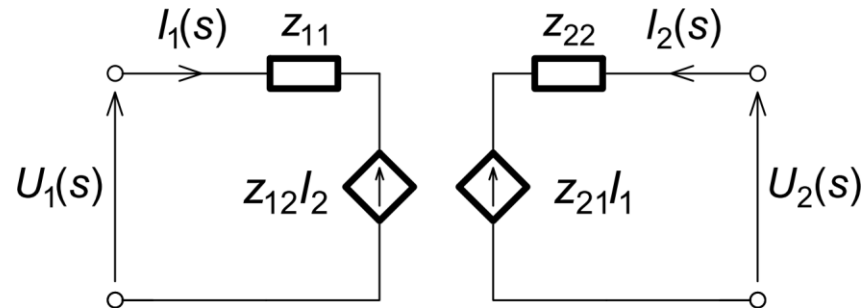
Two-Port Networks

OPERATOR CHARACTERIZATION

Impedance characterization

$$\begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix} = \begin{bmatrix} z_{11}(s) & z_{12}(s) \\ z_{21}(s) & z_{22}(s) \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \mathbf{Z}(s) \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix}$$

$$\mathbf{Z}(s) = \begin{bmatrix} z_{11}(s) & z_{12}(s) \\ z_{21}(s) & z_{22}(s) \end{bmatrix} \text{ - impedance matrix of two-port network}$$



General model for a two-port network characterized by $\mathbf{Z}$ matrix

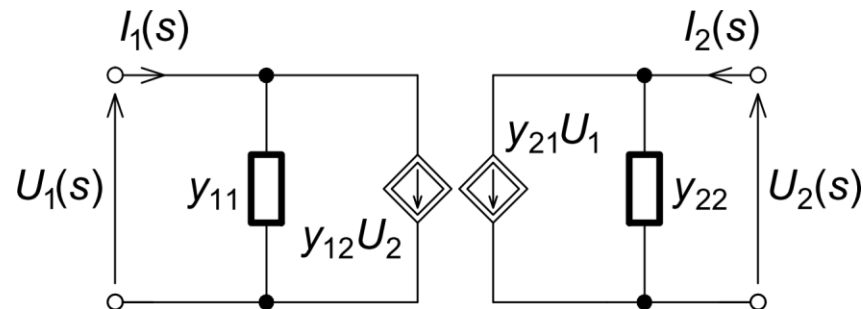
Two-Port Networks

OPERATOR CHARACTERIZATION

Admittance characterization

$$\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} y_{11}(s) & y_{12}(s) \\ y_{21}(s) & y_{22}(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix} = \mathbf{Y}(s) \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix}$$

$$\mathbf{Y}(s) = \begin{bmatrix} y_{11}(s) & y_{12}(s) \\ y_{21}(s) & y_{22}(s) \end{bmatrix} \text{ - admittance matrix of two-port network}$$



General model for a two-port network characterized by $\mathbf{Y}$ matrix

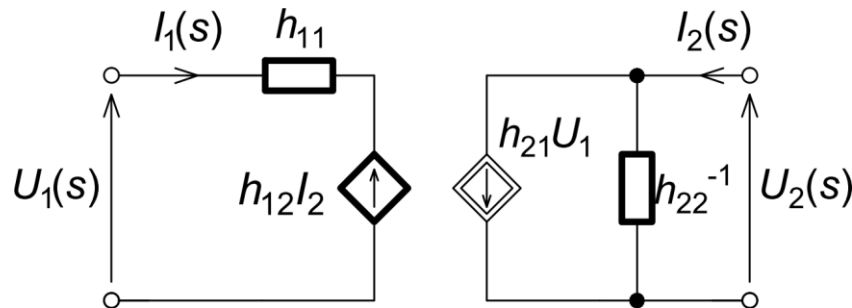
Two-Port Networks

OPERATOR CHARACTERIZATION

Hybrid characterization

$$\begin{bmatrix} U_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} h_{11}(s) & h_{12}(s) \\ h_{21}(s) & h_{22}(s) \end{bmatrix} \begin{bmatrix} I_1(s) \\ U_2(s) \end{bmatrix} = \mathbf{H}(s) \begin{bmatrix} I_1(s) \\ U_2(s) \end{bmatrix}$$

$$\mathbf{H}(s) = \begin{bmatrix} h_{11}(s) & h_{12}(s) \\ h_{21}(s) & h_{22}(s) \end{bmatrix} - \text{hybrid matrix of two-port network}$$



General model for a two-port network characterized by $\mathbf{H}$ matrix

Two-Port Networks

OPERATOR CHARACTERIZATION

Chain characterization

$$\begin{bmatrix} U_1(s) \\ I_1(s) \end{bmatrix} = \begin{bmatrix} a_{11}(s) & a_{12}(s) \\ a_{21}(s) & a_{22}(s) \end{bmatrix} \begin{bmatrix} U_2(s) \\ -I_2(s) \end{bmatrix} = \mathbf{A}(s) \begin{bmatrix} U_2(s) \\ -I_2(s) \end{bmatrix}$$

$$\mathbf{A}(s) = \begin{bmatrix} a_{11}(s) & a_{12}(s) \\ a_{21}(s) & a_{22}(s) \end{bmatrix} - \text{chain matrix of two-port network}$$

$$\mathbf{A}(s) = \begin{bmatrix} A(s) & B(s) \\ C(s) & D(s) \end{bmatrix} - \text{ABCD matrix (alternatively used form)}$$

Elements of matrices $\mathbf{Z}(s)$, $\mathbf{Y}(s)$, $\mathbf{H}(s)$, and $\mathbf{A}(s)$ are impedance, admittance, hybrid, and chain (ABCD) parameters, respectively. They are called **characteristic parameters of two-port network**. They do not depend on external circuits connected to the two-port network.

Two-Port Networks

CHARACTERIZATION IN COMPLEX DOMAIN

If a two-port network (including connected circuits) is BIBO stable and all excitations are sinusoidal with the same ω_0 , then for steady state we can introduce characterization in complex domain.

Impedance characterization

$$\begin{bmatrix} \underline{U}_1 \\ \underline{U}_2 \end{bmatrix} = \begin{bmatrix} \underline{Z}_{11} & \underline{Z}_{12} \\ \underline{Z}_{21} & \underline{Z}_{22} \end{bmatrix} \begin{bmatrix} \underline{I}_1 \\ \underline{I}_2 \end{bmatrix} = \underline{\mathbf{Z}} \begin{bmatrix} \underline{I}_1 \\ \underline{I}_2 \end{bmatrix} \quad \underline{\mathbf{Z}} = \mathbf{Z}(s) \Big|_{s = j\omega_0}$$

Admittance characterization

$$\begin{bmatrix} \underline{I}_1 \\ \underline{I}_2 \end{bmatrix} = \begin{bmatrix} \underline{y}_{11} & \underline{y}_{12} \\ \underline{y}_{21} & \underline{y}_{22} \end{bmatrix} \begin{bmatrix} \underline{U}_1 \\ \underline{U}_2 \end{bmatrix} = \underline{\mathbf{Y}} \begin{bmatrix} \underline{U}_1 \\ \underline{U}_2 \end{bmatrix} \quad \underline{\mathbf{Y}} = \mathbf{Y}(s) \Big|_{s = j\omega_0}$$

Chain characterization

$$\begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix} = \begin{bmatrix} \underline{a}_{11} & \underline{a}_{12} \\ \underline{a}_{21} & \underline{a}_{22} \end{bmatrix} \begin{bmatrix} \underline{U}_2 \\ -\underline{I}_2 \end{bmatrix} = \underline{\mathbf{A}} \begin{bmatrix} \underline{U}_2 \\ -\underline{I}_2 \end{bmatrix} \quad \underline{\mathbf{A}} = \mathbf{A}(s) \Big|_{s = j\omega_0}$$

Two-Port Networks

CHANGE OF MATRIX CHARACTERIZATION

Circuit's characterization with a given matrix can be exchanged for any other matrix according to the table below.

	$\mathbf{Z}$	$\mathbf{Y}$	$\mathbf{A}$
$\mathbf{Z}$		$\begin{bmatrix} \frac{z_{22}}{\det \mathbf{Z}} & \frac{-z_{12}}{\det \mathbf{Z}} \\ \frac{-z_{21}}{\det \mathbf{Z}} & \frac{z_{11}}{\det \mathbf{Z}} \end{bmatrix}$	$\begin{bmatrix} \frac{z_{11}}{z_{21}} & \frac{\det \mathbf{Z}}{z_{21}} \\ 1 & \frac{z_{22}}{z_{21}} \end{bmatrix}$
$\mathbf{Y}$	$\begin{bmatrix} \frac{y_{22}}{\det \mathbf{Y}} & \frac{-y_{12}}{\det \mathbf{Y}} \\ \frac{-y_{21}}{\det \mathbf{Y}} & \frac{y_{11}}{\det \mathbf{Y}} \end{bmatrix}$		$\begin{bmatrix} -\frac{y_{22}}{y_{21}} & -\frac{1}{y_{21}} \\ \det \mathbf{Y} & \frac{y_{11}}{y_{21}} \end{bmatrix}$
$\mathbf{A}$	$\begin{bmatrix} \frac{a_{11}}{a_{21}} & \frac{\det \mathbf{A}}{a_{21}} \\ 1 & \frac{a_{22}}{a_{21}} \end{bmatrix}$	$\begin{bmatrix} \frac{a_{22}}{a_{12}} & -\frac{\det \mathbf{A}}{a_{12}} \\ -\frac{1}{a_{12}} & \frac{a_{11}}{a_{12}} \end{bmatrix}$	

Characterization other matrices such as with $\mathbf{G}$ matrix (inversed $\mathbf{H}$), $\mathbf{S}$ matrix (scattering matrix), $\mathbf{T}$ (scattering transfer matrix) can also be used.

Two-Port Networks

BASIC PROPERTIES

Reciprocity

A network is said to be reciprocal if the voltage appearing at port 2 due to a current applied at port 1 is the same as the voltage appearing at port 1 when the same current is applied to port 2.

A network that consists entirely of linear passive components (that is, resistors, capacitors and inductors) is usually reciprocal, a notable exception being passive *circulators* and *isolators* that contain magnetized materials. In general, it *will not* be reciprocal if it contains active components.

Condition for **Z** matrix

$$Z_{21} = Z_{12}$$

Condition for **Y** matrix

$$y_{21} = y_{12}$$

Condition for **A** matrix

$$\det \mathbf{A} = 1$$

Two-Port Networks

BASIC PROPERTIES

Symmetry

A network is said to be symmetrical if the impedance measured at one port is equal to the impedance measured at the other port with remaining port open circuited.

Condition for **Z** matrix

$$Z_{11} = Z_{22}$$

Condition for **Y** matrix

$$y_{11} = y_{22}$$

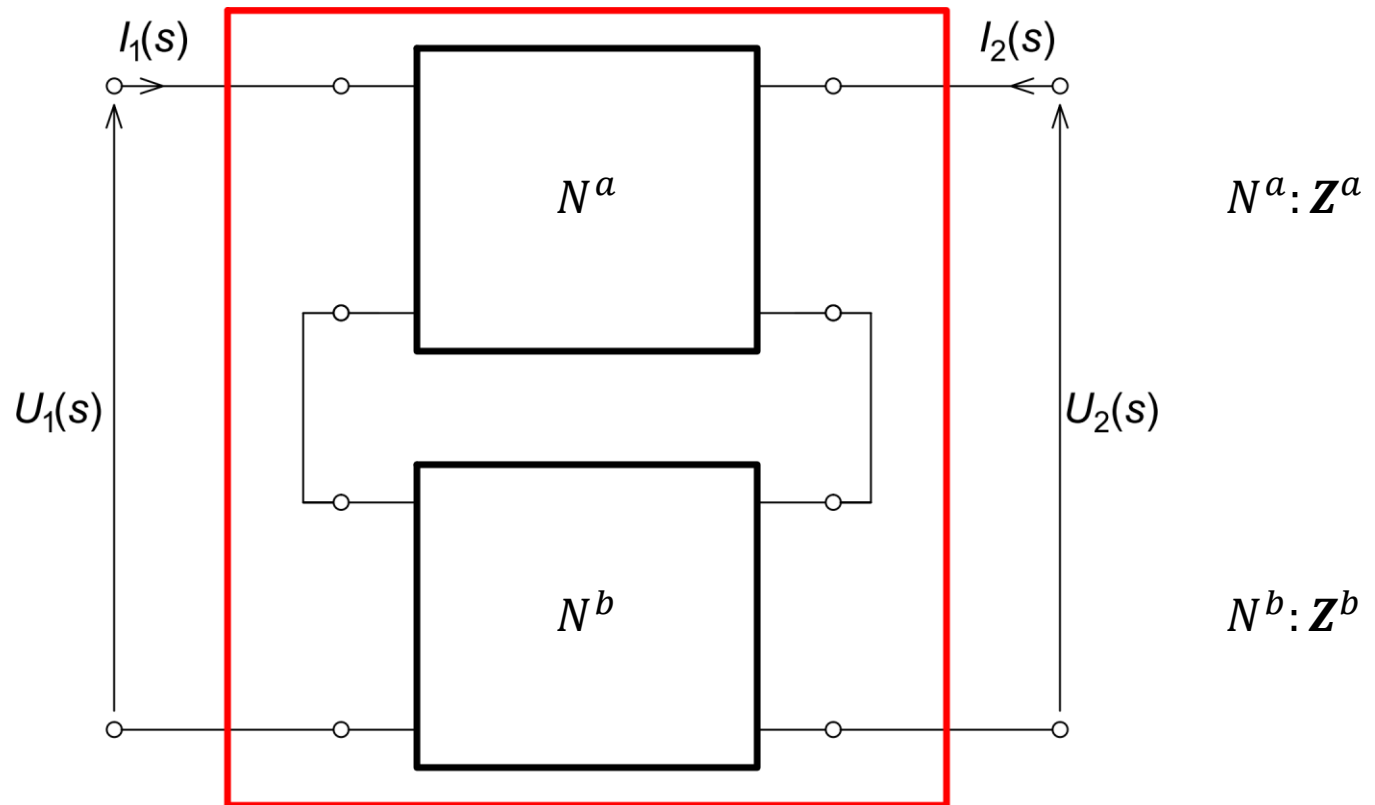
Condition for **A** matrix

$$a_{11} = a_{22}$$

Two-Port Networks

CONNECTIONS OF TWO-PORT NETWORKS

Series connection

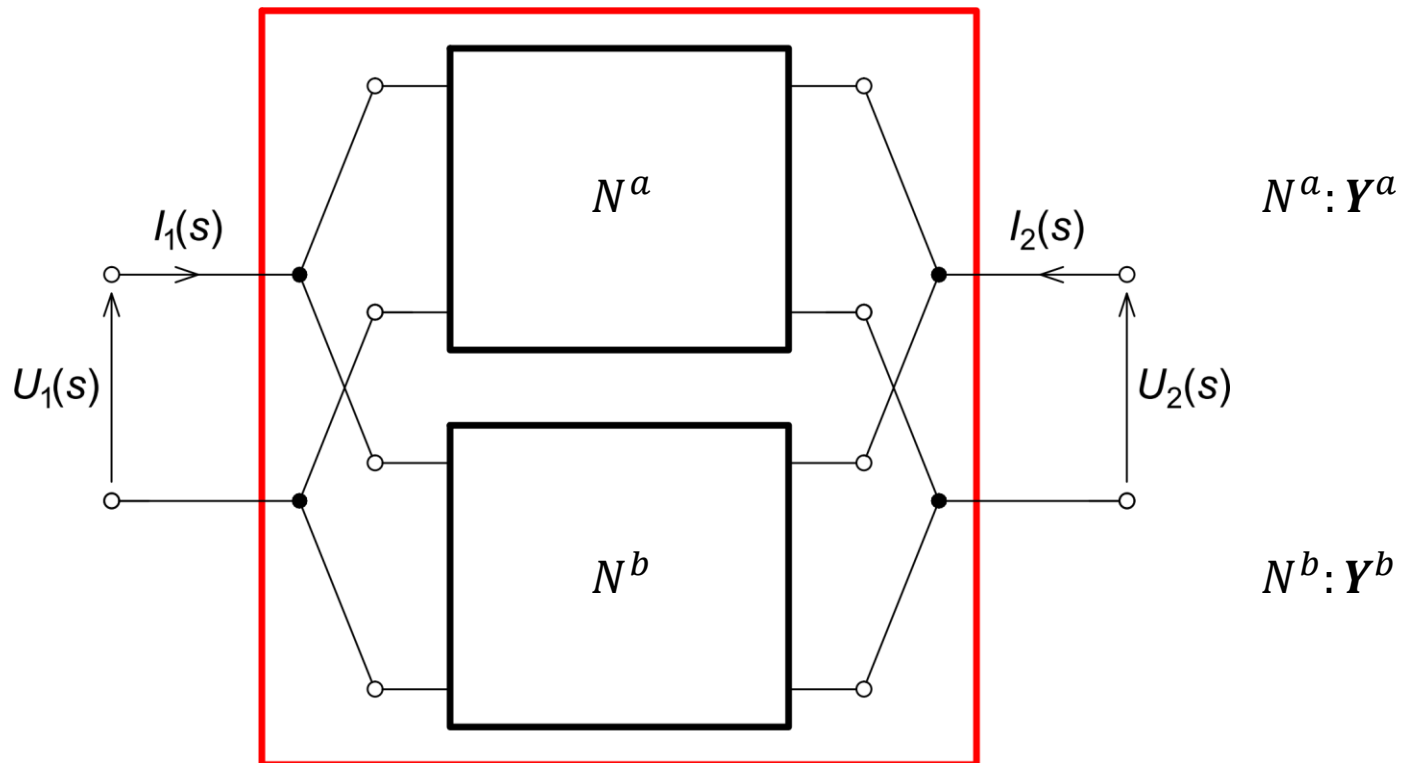


$$N: \mathbf{Z} = \mathbf{Z}^a + \mathbf{Z}^b$$

Two-Port Networks

CONNECTIONS OF TWO-PORT NETWORKS

Parallel connection

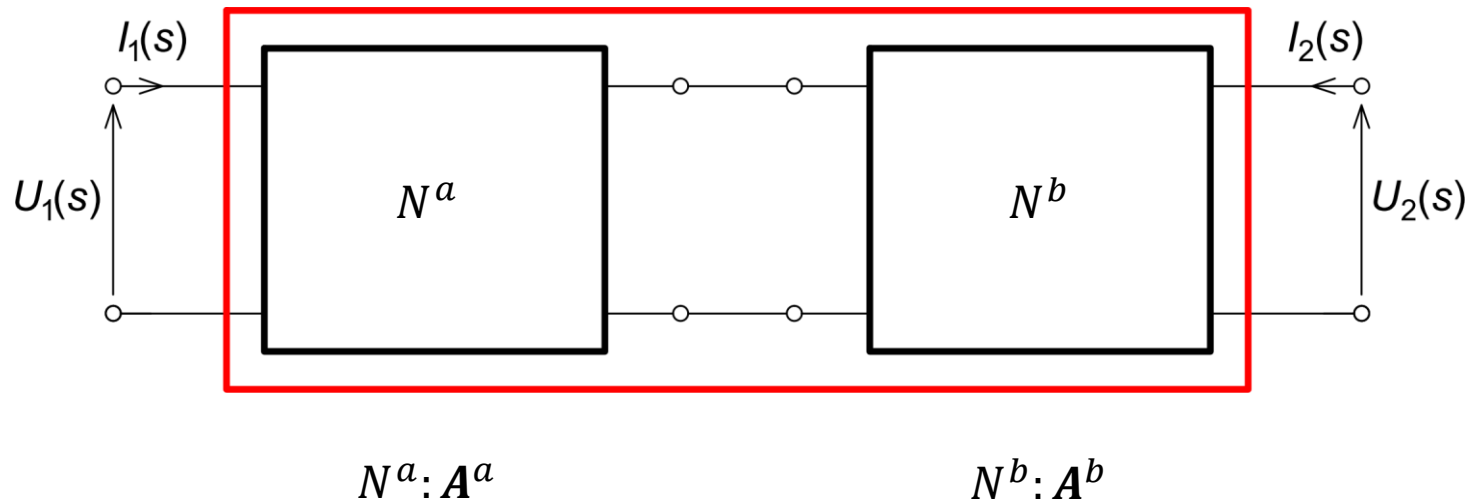


$$N: Y = Y^a + Y^b$$

Two-Port Networks

CONNECTIONS OF TWO-PORT NETWORKS

Cascade (chain) connection



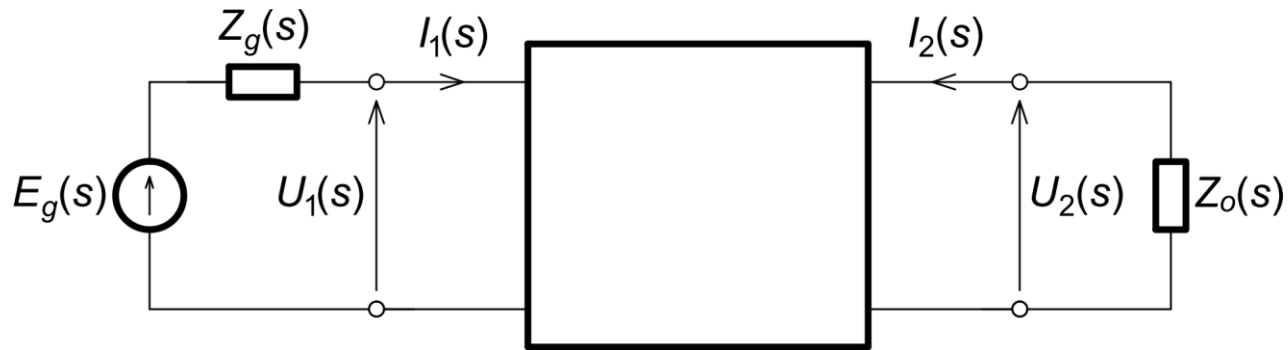
$$N: \mathbf{A} = \mathbf{A}^a \mathbf{A}^b$$

Please note that:

$$\mathbf{A}^a \mathbf{A}^b \neq \mathbf{A}^b \mathbf{A}^a$$

Two-Port Networks

WORKING PARAMETERS



$$K_i(s) \triangleq \frac{I_2(s)}{I_1(s)} \text{ - current amplification}$$

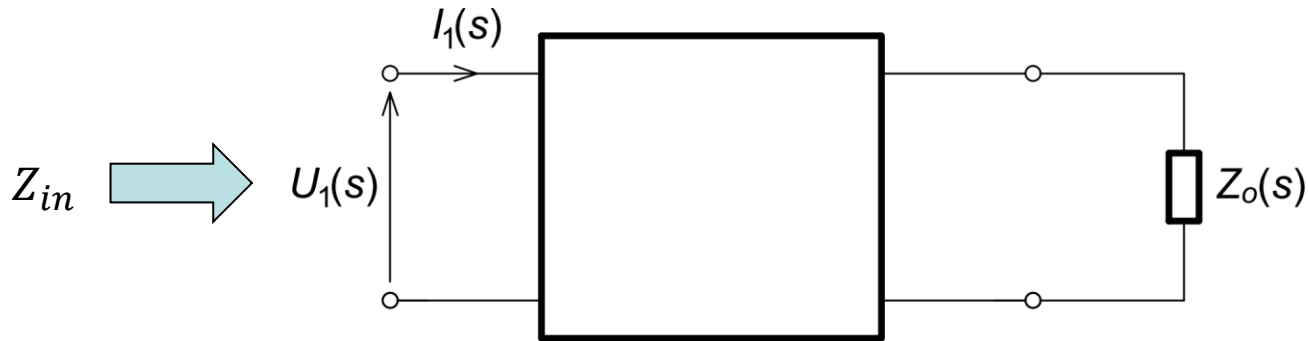
$$K_u(s) \triangleq \frac{U_2(s)}{U_1(s)} \text{ -voltage amplification}$$

$$K_{uRMS}(s) \triangleq \frac{U_2(s)}{E_g(s)} \text{ - RMS voltage amplification}$$

Two-Port Networks

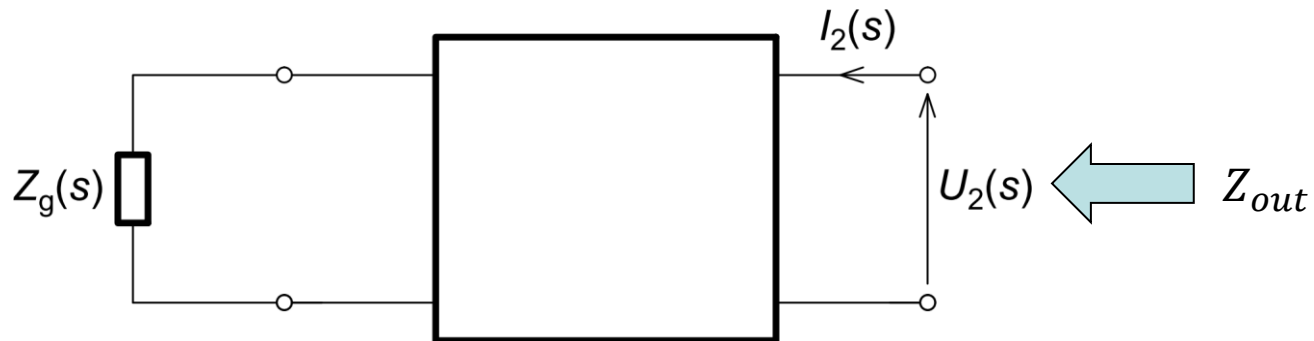
WORKING PARAMETERS

Input impedance



$$Z_{in}(s) = \frac{U_1(s)}{I_1(s)}$$

Output impedance



$$Z_{out}(s) = \frac{U_2(s)}{I_2(s)}$$

Two-Port Networks

WORKING PARAMETERS

	$\mathbf{Z}(s)$
$Z_{in}(s)$	$\frac{z_{11}(s)Z_o(s) + \det\mathbf{Z}(s)}{z_{22}(s) + Z_o(s)}$
$Z_{out}(s)$	$\frac{z_{22}(s)Z_g(s) + \det\mathbf{Z}(s)}{z_{11}(s) + Z_g(s)}$
$K_i(s)$	$-\frac{z_{21}(s)}{z_{22}(s) + Z_g(s)}$
$K_u(s)$	$\frac{z_{21}(s)Z_o(s)}{z_{11}(s)Z_o(s) + \det\mathbf{Z}(s)}$
$K_{uRMS}(s)$	$\frac{z_{21}(s)Z_o(s)}{z_{11}(s)Z_o(s) + z_{22}(s)Z_g(s) + Z_o(s)Z_g(s) + \det\mathbf{Z}(s)}$

Two-Port Networks

WORKING PARAMETERS

	$Y(s)$
$Z_{in}(s)$	$\frac{1 + y_{22}(s)Z_o(s)}{y_{11}(s) + Z_o(s)\det Y(s)}$
$Z_{out}(s)$	$\frac{1 + y_{11}(s)Z_g(s)}{y_{22}(s) + Z_g(s)\det Y(s)}$
$K_i(s)$	$\frac{y_{21}(s)}{y_{22}(s) + Z_g(s)\det Y(s)}$
$K_u(s)$	$-\frac{y_{21}(s)Z_o(s)}{1 + y_{22}(s)Z_o(s)}$
$K_{uRMS}(s)$	$\frac{y_{21}(s)Z_o(s)}{1 + y_{11}(s)Z_g(s) + y_{22}(s)Z_o(s) + Z_g(s)Z_o(s)\det Y(s)}$

Two-Port Networks

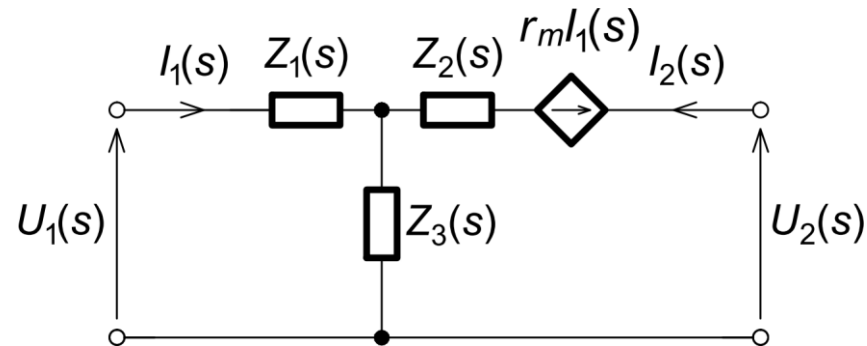
WORKING PARAMETERS

	$A(s)$
$Z_{in}(s)$	$\frac{a_{11}(s)Z_o(s) + a_{12}(s)}{a_{21}(s)Z_o(s) + a_{22}(s)}$
$Z_{out}(s)$	$\frac{a_{22}(s)Z_g(s) + a_{12}(s)}{a_{21}(s)Z_g(s) + a_{11}(s)}$
$K_i(s)$	$\frac{1}{a_{21}(s)Z_o(s) + a_{22}(s)}$
$K_u(s)$	$\frac{Z_o(s)}{a_{11}(s)Z_o(s) + a_{12}(s)}$
$K_{uRMS}(s)$	$\frac{Z_0(s)}{a_{12}(s) + a_{11}(s)Z_o(s) + a_{22}(s)Z_g(s) + a_{21}(s)Z_o(s)Z_g(s)}$

Two-Port Networks

EXAMPLE 1

Find impedance parameters of a T-shape two-port network shown below:

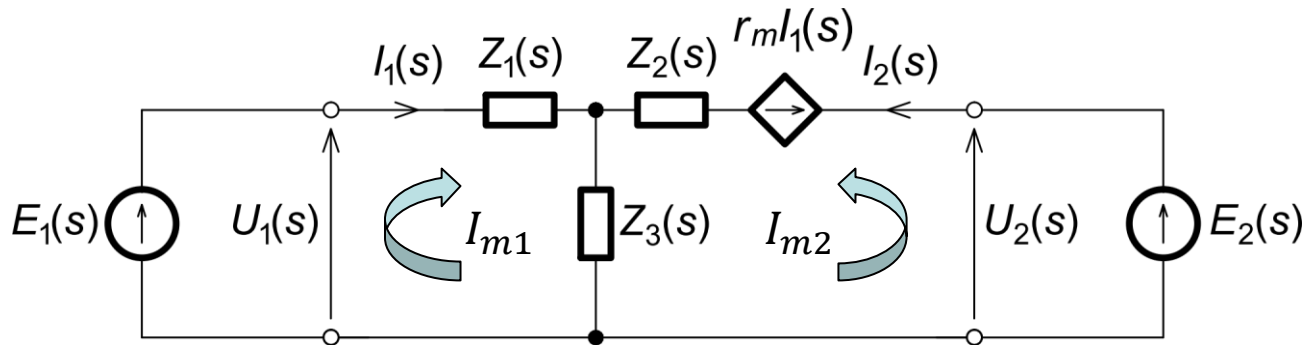


Two-Port Networks

EXAMPLE 1

Solution:

After connecting voltage sources to the two-port network's ports it is possible to write equations for loop currents choosing independent loops.



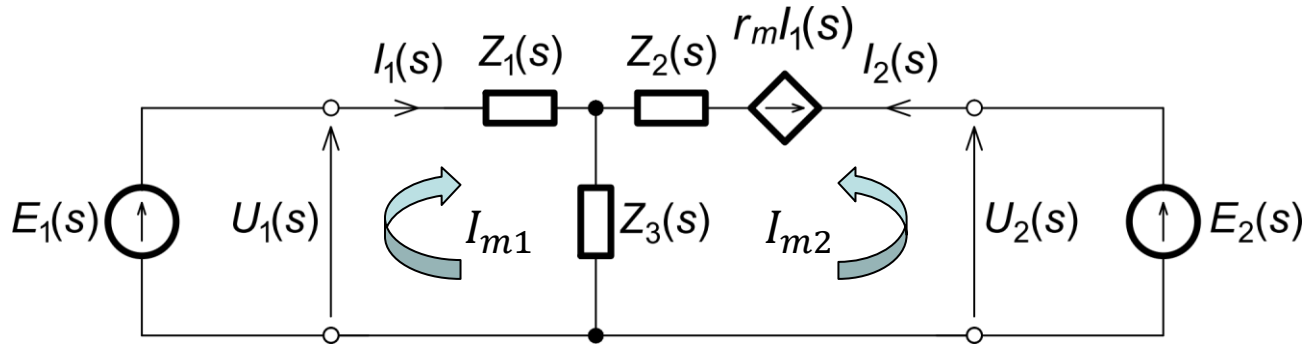
We get following set of equations for loop currents:

$$\begin{bmatrix} Z_1(s) + Z_3(s) & Z_3(s) \\ Z_3(s) + r_m & Z_2(s) + Z_3(s) \end{bmatrix} \begin{bmatrix} I_{m1}(s) \\ I_{m2}(s) \end{bmatrix} = \begin{bmatrix} E_1(s) \\ E_2(s) \end{bmatrix}$$

Note, that it is convenient to choose directions of loop currents differently than for classic mesh method.

Two-Port Networks

EXAMPLE 1



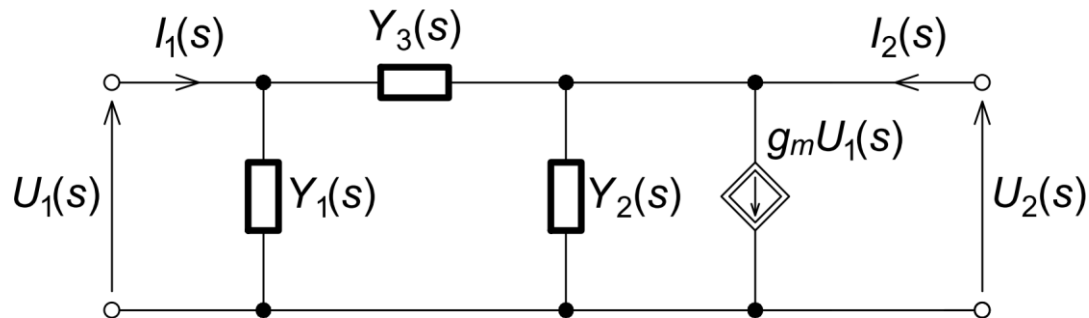
By including that $U_1(s) = E_1(s)$, $U_2(s) = E_2(s)$ and $I_1(s) = I_{m1}(s)$, $I_2(s) = I_{m2}(s)$ we can easily see that these equations form impedance matrix $\mathbf{Z}$ of the two-port network:

$$\mathbf{Z}(s) = \begin{bmatrix} Z_1(s) + Z_3(s) & Z_3(s) \\ Z_3(s) + r_m & Z_2(s) + Z_3(s) \end{bmatrix}$$

Two-Port Networks

EXAMPLE 2

Find admittance parameters of a Π -shape two-port network shown below:

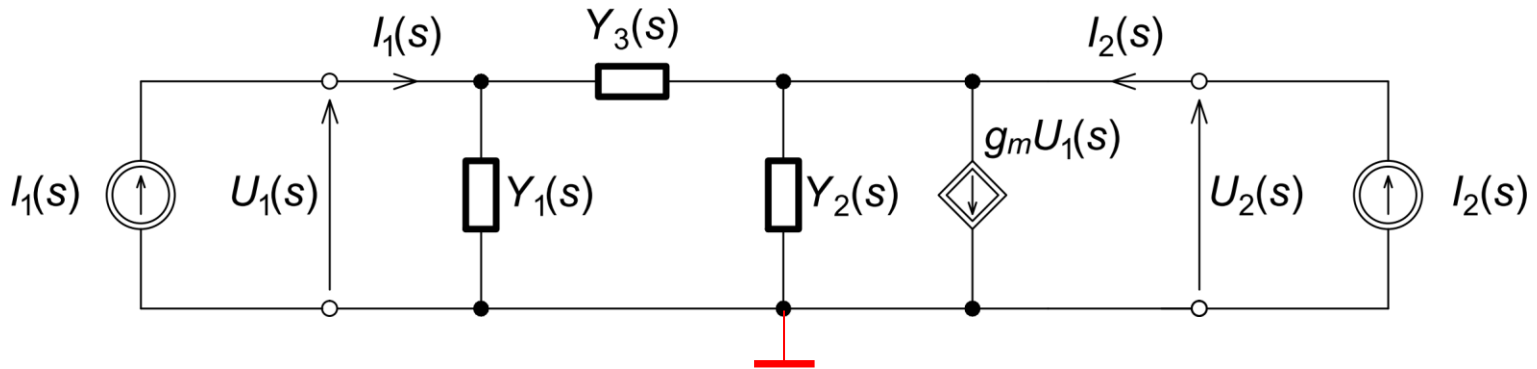


Two-Port Networks

EXAMPLE 2

Solution:

We apply method similar to example 1. This time we connect current sources to the ports of two-port network and write equations for node-voltage analysis:



$$\begin{bmatrix} Y_1(s) + Y_3(s) & -Y_3(s) \\ -Y_3(s) + g_m & Y_2(s) + Y_3(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix} = \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix}$$

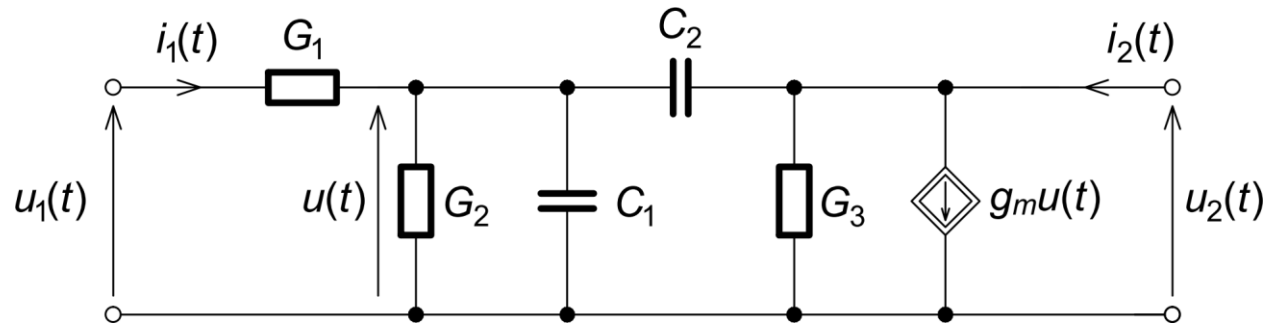
Therefore:

$$\mathbf{Y}(s) = \begin{bmatrix} Y_1(s) + Y_3(s) & -Y_3(s) \\ -Y_3(s) + g_m & Y_2(s) + Y_3(s) \end{bmatrix}$$

Two-Port Networks

EXAMPLE 3

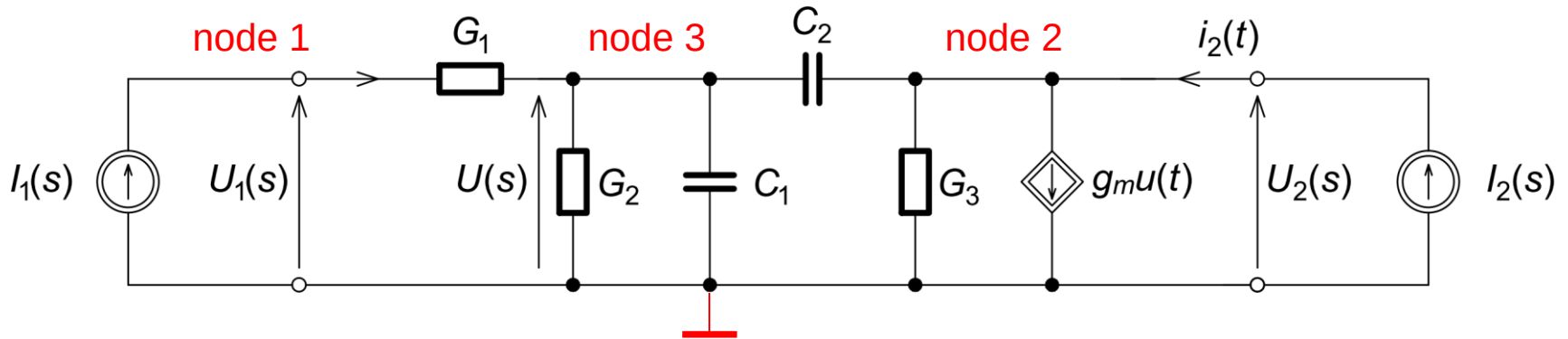
Find admittance parameters of a two-port network shown below:



Two-Port Networks

EXAMPLE 3

After connecting current sources to ports of two-port network a new circuit is obtained:



With denotation of nodes the circuit can be represented with equations for node-voltage analysis:

1. $G_1 U_1(s) - G_1 U_3 = I_1(s)$
2. $(sC_2 + G_3)U_2(s) - (sC_2 - g_m)U_3(s) = I_2(s),$
3. $-G_1 U_1(s) - sC_2 U_2(s) + (sC_1 + sC_2 + G_1 + G_2)U_3(s) = 0$

From equation 3 we get:

$$U_3(s) = \frac{G_1}{sC_1 + sC_2 + G_1 + G_2} U_1(s) + \frac{sC_2}{sC_1 + sC_2 + G_1 + G_2} U_2(s)$$

Two-Port Networks

EXAMPLE 3

1. $G_1 U_1(s) - G_1 U_3 = I_1(s)$
2. $(sC_2 + G_3)U_2(s) - (sC_2 - g_m)U_3(s) = I_2(s),$
3. $-G_1 U_1(s) - sC_2 U_2(s) + (sC_1 + sC_2 + G_1 + G_2)U_3(s) = 0$

From equation 3 we get:

$$U_3(s) = \frac{G_1}{sC_1 + sC_2 + G_1 + G_2} U_1(s) + \frac{sC_2}{sC_1 + sC_2 + G_1 + G_2} U_2(s)$$

Substituting this formula to equations 1 and 2 we get:

$$\begin{aligned} \left(G_1 - \frac{G_1^2}{sC_1 + sC_2 + G_1 + G_2} \right) U_1(s) - \frac{sC_2 G_1}{sC_1 + sC_2 + G_1 + G_2} U_2(s) &= I_1(s) \\ -\frac{(sC_2 - g_m)G_1}{sC_1 + sC_2 + G_1 + G_2} U_1(s) + \left[sC_2 + G_3 - \frac{sC_2(sC_2 - g_m)}{sC_1 + sC_2 + G_1 + G_2} \right] U_2(s) &= I_2(s) \end{aligned}$$

Two-Port Networks

EXAMPLE 3

Substituting this formula to equations 1 and 2 we get:

$$\left(G_1 - \frac{G_1^2}{sC_1 + sC_2 + G_1 + G_2} \right) U_1(s) - \frac{sC_2 G_1}{sC_1 + sC_2 + G_1 + G_2} U_2(s) = I_1(s)$$

$$- \frac{(sC_2 - g_m)G_1}{sC_1 + sC_2 + G_1 + G_2} U_1(s) + \left[sC_2 + G_3 - \frac{sC_2(sC_2 - g_m)}{sC_1 + sC_2 + G_1 + G_2} \right] U_2(s) = I_2(s)$$

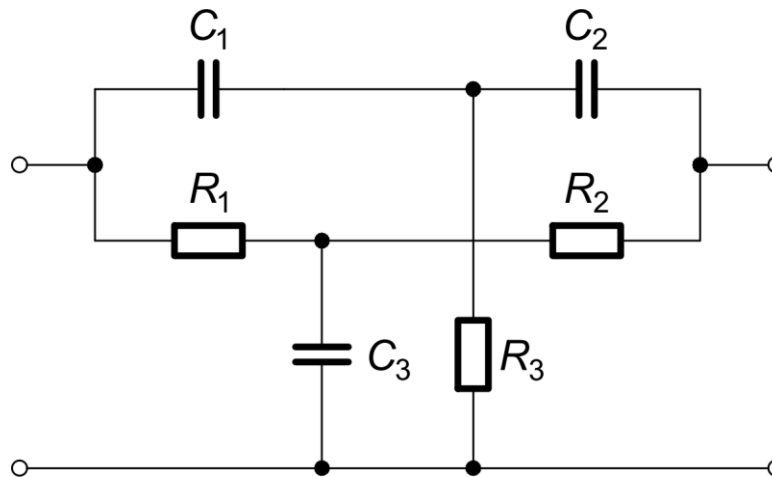
from which the admittance matrix can be derived:

$$\mathbf{Y} = \begin{bmatrix} G_1 - \frac{G_1^2}{sC_1 + sC_2 + G_1 + G_2} & - \frac{sC_2 G_1}{sC_1 + sC_2 + G_1 + G_2} \\ - \frac{(sC_2 - g_m)G_1}{sC_1 + sC_2 + G_1 + G_2} & sC_2 + G_3 - \frac{sC_2(sC_2 - g_m)}{sC_1 + sC_2 + G_1 + G_2} \end{bmatrix}$$

Two-Port Networks

EXAMPLE 4

Find operator and complex parameters of a two-port network shown below:



$$R_1 = R_2 = 1\Omega, \quad R_3 = \frac{1}{2}\Omega$$

$$C_1 = C_2 = 1F, \quad C_3 = 2F$$

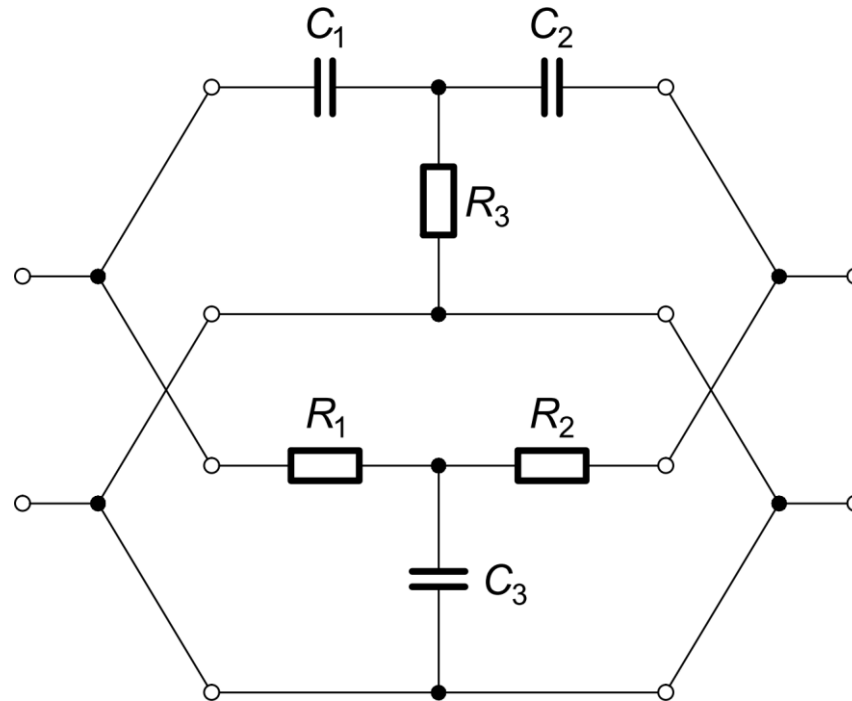
$$\omega_0 = 1 \frac{rad}{s}$$

Two-Port Networks

EXAMPLE 4

Solution:

The considered two-port network can be analyzed as a parallel connection of two simpler two-port networks:



These two-port networks have simple T-type structure (finding impedance parameters of that kind of two-port network was goal in example 1). The solution of this example can be brought down to finding admittance parameters $\mathbf{Y}_1$ and $\mathbf{Y}_2$ based on impedance parameters $\mathbf{Z}_1$ and $\mathbf{Z}_2$, and adding two calculated matrices.

Two-Port Networks

EXAMPLE 4

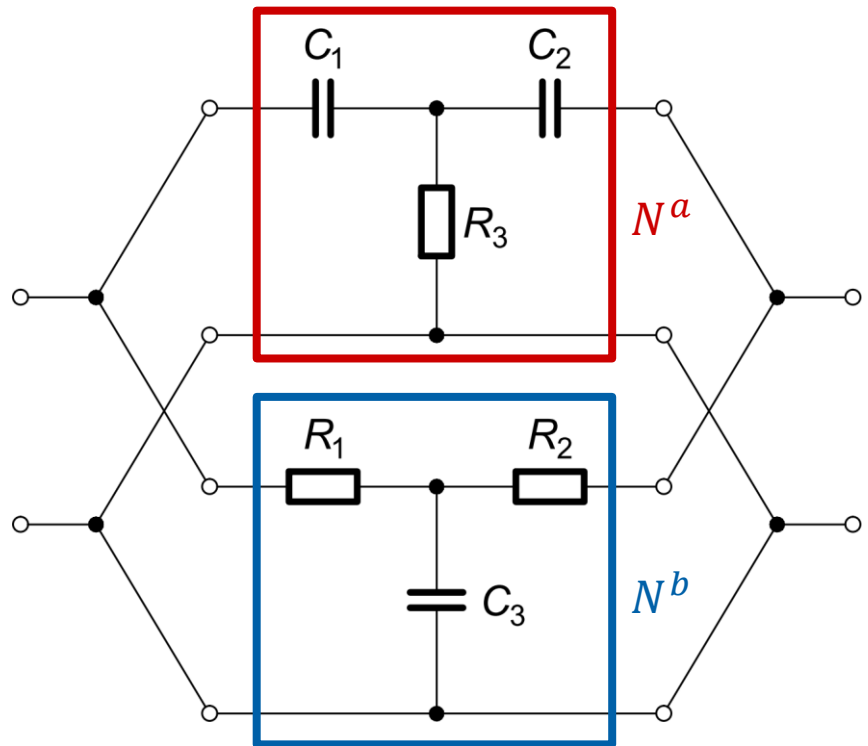
$$N^a: \mathbf{Z}^a(s) = \begin{bmatrix} R_3 + \frac{1}{sC_1} & R_3 \\ R_3 & R_3 + \frac{1}{sC_2} \end{bmatrix}$$

$$N^b: \mathbf{Z}^b(s) = \begin{bmatrix} R_1 + \frac{1}{sC_3} & \frac{1}{sC_3} \\ \frac{1}{sC_3} & R_2 + \frac{1}{sC_3} \end{bmatrix}$$

We calculate, respectively:

$$\det \mathbf{Z}^a(s) = \left(R_3 + \frac{1}{sC_1} \right) \left(R_3 + \frac{1}{sC_2} \right) - R_3^2 = \frac{s(C_1 + C_2)R_3 + 1}{s^2 C_1 C_2}$$

$$\det \mathbf{Z}^b(s) = \left(R_1 + \frac{1}{sC_3} \right) \left(R_2 + \frac{1}{sC_3} \right) - \frac{1}{s^2 C_3^2} = \frac{sC_3 R_1 R_2 + R_1 + R_2}{sC_3}$$



Two-Port Networks

EXAMPLE 4

and then:

$$\mathbf{Y}^a(s) = [\mathbf{Z}^a(s)]^{-1} = \begin{bmatrix} \frac{s^2 C_1 C_2 R_3 + s C_1}{s(C_1 + C_2)R_3 + 1} & -\frac{s^2 C_1 C_2 R_3}{s(C_1 + C_2)R_3 + 1} \\ -\frac{s^2 C_1 C_2 R_3}{s(C_1 + C_2)R_3 + 1} & \frac{s^2 C_1 C_2 R_3 + s C_2}{s(C_1 + C_2)R_3 + 1} \end{bmatrix} = \begin{bmatrix} \frac{s(s+2)}{2(s+1)} & -\frac{s^2}{2(s+1)} \\ -\frac{s^2}{2(s+1)} & \frac{s(s+2)}{1(s+1)} \end{bmatrix}$$

$$\mathbf{Y}^b(s) = [\mathbf{Z}^b(s)]^{-1} = \begin{bmatrix} \frac{s C_3 R_2 + 1}{s C_3 R_1 R_2 + R_1 + R_2} & -\frac{1}{s C_3 R_1 R_2 + R_1 + R_2} \\ \frac{1}{s C_3 R_1 R_2 + R_1 + R_2} & \frac{s C_3 R_1 + 1}{s C_3 R_1 R_2 + R_1 + R_2} \end{bmatrix}$$

$$\mathbf{Y}^b(s) = [\mathbf{Z}^b(s)]^{-1} = \begin{bmatrix} \frac{2s + 1}{2(s + 1)} & -\frac{1}{2(s + 1)} \\ -\frac{1}{2(s + 1)} & \frac{2s + 1}{2(s + 1)} \end{bmatrix}$$

Two-Port Networks

EXAMPLE 4

Finally:

$$\mathbf{Y}(s) = \mathbf{Y}^a(s) + \mathbf{Y}^b(s) = \begin{bmatrix} \frac{s^2 + 4s + 1}{2(s + 1)} & -\frac{s^2 + 1}{2(s + 1)} \\ -\frac{s^2 + 1}{2(s + 1)} & \frac{s^2 + 4s + 1}{2(s + 1)} \end{bmatrix}$$

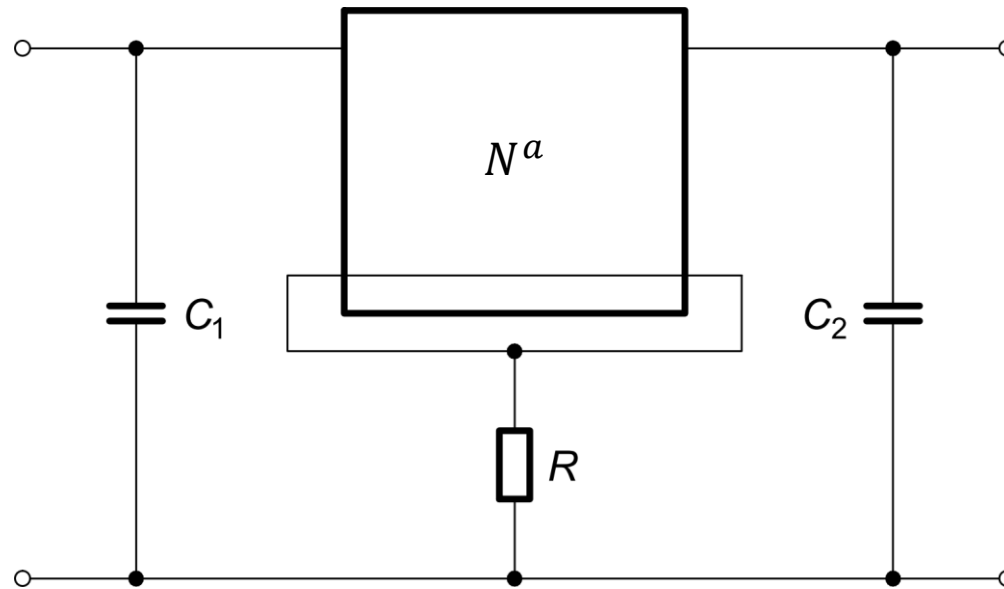
Complex admittance matrix for sinusoidal excitation with $\omega_0 = 1$ can be obtained by substituting $s = j$. Then:

$$\mathbf{Y} = \begin{bmatrix} 1 + j & 0 \\ 0 & 1 + j \end{bmatrix}$$

Two-Port Networks

EXAMPLE 5

Find complex parameters of a two-port network circuit shown below:



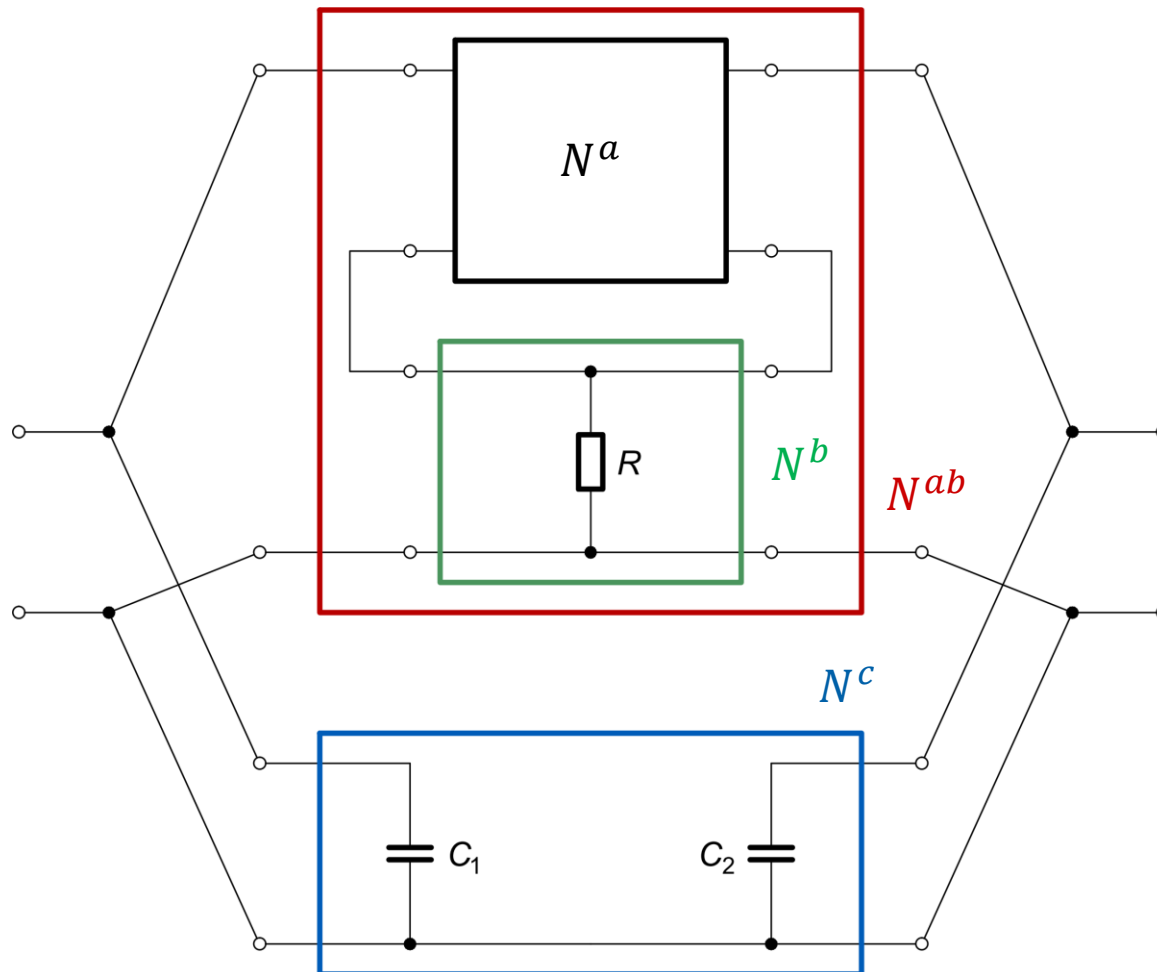
$$N^a: \mathbf{A}^a = \begin{bmatrix} 1 + j2 & 2 \\ j & 1 \end{bmatrix}$$

$$R = 1\Omega, C_1 = 1F, C_2 = \frac{1}{2}F, \omega_0 = 1 \frac{\text{rad}}{\text{s}}$$

Two-Port Networks

EXAMPLE 5

The circuit can be decomposed into three simpler two-ports appropriately connected:



Two-Port Networks

EXAMPLE 5

We can calculate:

$$N^a: \mathbf{A}^a \rightarrow \mathbf{Z}^a = \begin{bmatrix} 2-j & -j \\ -j & -j \end{bmatrix}$$

$$N^b: \mathbf{Z}^b = \begin{bmatrix} R & R \\ R & R \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$N^{ab}: \mathbf{Z}^{ab} = \mathbf{Z}^a + \mathbf{Z}^b = \begin{bmatrix} 3-j & 1-j \\ 1-j & 1-j \end{bmatrix} \rightarrow \mathbf{Y}^{ab} = \mathbf{Z}^{ab^{-1}} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 + \frac{1}{2}j \end{bmatrix}$$

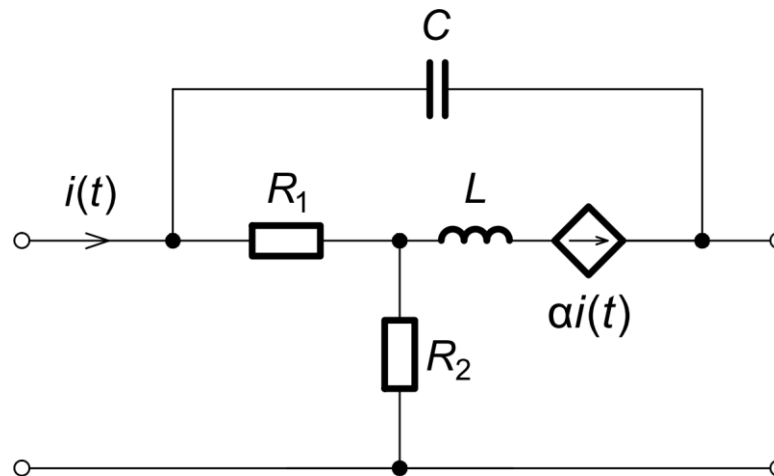
$$N^c: \mathbf{Y}^c = \begin{bmatrix} j\omega_0 & 0 \\ 0 & j\omega_0 \end{bmatrix} = \begin{bmatrix} j & 0 \\ 0 & j\frac{1}{2} \end{bmatrix}$$

$$\mathbf{Y} = \mathbf{Y}^{ab} + \mathbf{Y}^c = \begin{bmatrix} \frac{1}{2} + j & -\frac{1}{2} \\ -\frac{1}{2} & 1 + j \end{bmatrix}$$

Two-Port Networks

EXAMPLE 6

Find complex parameters of two-port network show on a drawing:

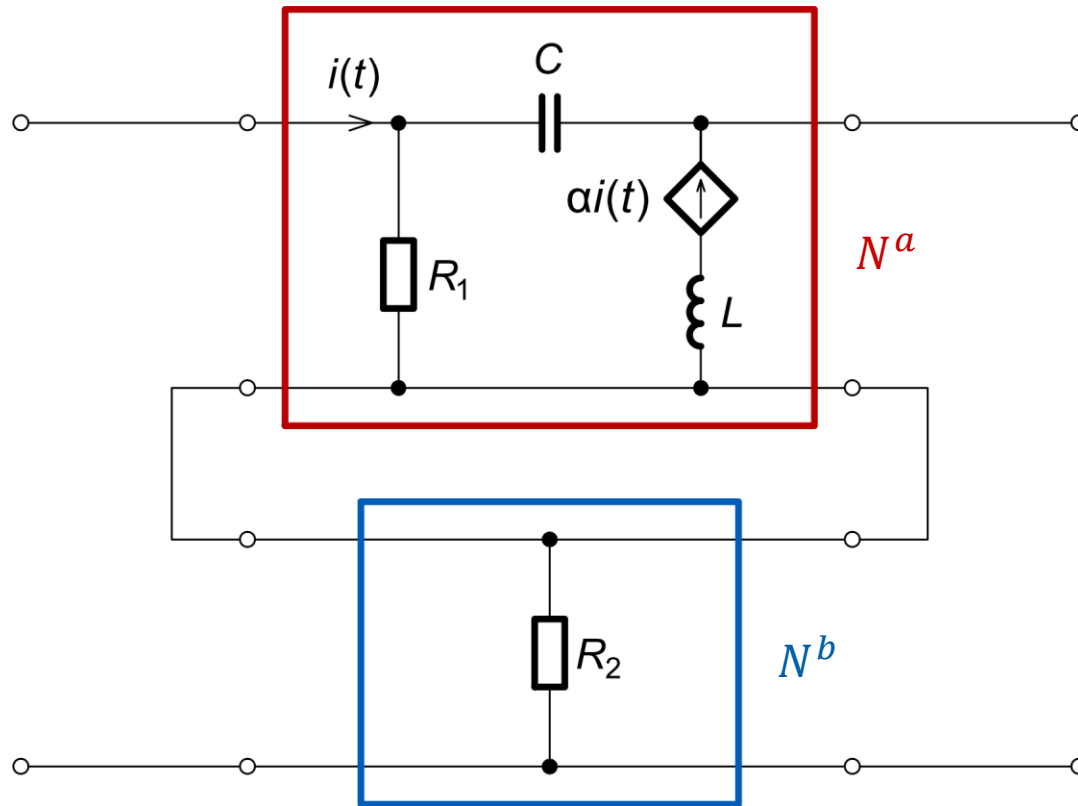


$$R_1 = 1\Omega, R_2 = 2\Omega, L = \frac{1}{2}H, C = 1F, \rho = 2\Omega, \omega_0 = 1\frac{rad}{s}$$

Two-Port Networks

EXAMPLE 6

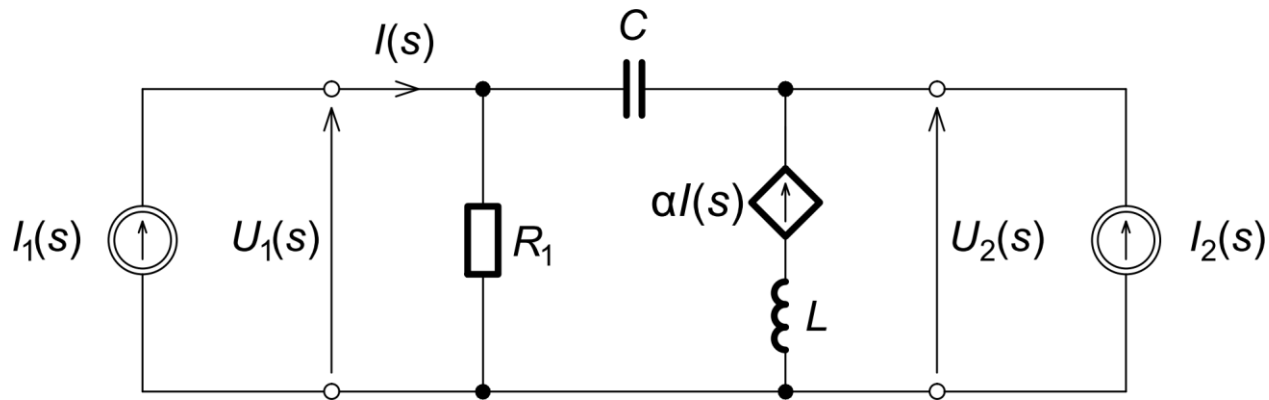
This two-port network can be decomposed as shown below:



Two-Port Networks

EXAMPLE 6

The two-port network can be considered as a series connection of two-port networks N^a and N^b . Their impedance matrices need to be found. The schematic diagram of the decomposed two-port with current sources connected to its ports is shown below:



By writing set of equations for node-voltage analysis we obtain admittance characterization of the two-port network:

$$\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} sC + \frac{1}{R_1} & -sC \\ -sC - \frac{\alpha C}{L} - \frac{\alpha}{sLR_1} & sC + \frac{\alpha C}{L} + \frac{1}{sL} \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix}$$

Two-Port Networks

EXAMPLE 6

After substituting numerical value we get:

$$\mathbf{Y}^a(s) = \begin{bmatrix} s + 1 & -s \\ -s - 4 - \frac{4}{s} & s + 4 + \frac{2}{s} \end{bmatrix}$$

Impedance matrix:

$$\mathbf{Z}^a(s) = [\mathbf{Y}^a(s)]^{-1} = \begin{bmatrix} \frac{s^2 + 4s + 2}{s^2 + 2s + 2} & \frac{s^2}{s^2 + 2s + 2} \\ \frac{s^2 + 4s + 4}{s^2 + 2s + 2} & \frac{s^2 + s}{s^2 + 2s + 2} \end{bmatrix}$$

Two-Port Networks

EXAMPLE 6

Impedance matrix of the two-port N^b :

$$\mathbf{Z}^b(s) = \begin{bmatrix} R_2 & R_2 \\ R_2 & R_2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Finally we get:

$$\mathbf{Z}(s) = \mathbf{Z}^a(s) + \mathbf{Z}^b(s) = \begin{bmatrix} \frac{3s^2 + 8s + 6}{s^2 + 2s + 2} & \frac{3s^2 + 4s + 4}{s^2 + 2s + 2} \\ \frac{3s^2 + 8s + 8}{s^2 + 2s + 2} & \frac{3s^2 + 5s + 4}{s^2 + 2s + 2} \end{bmatrix}$$

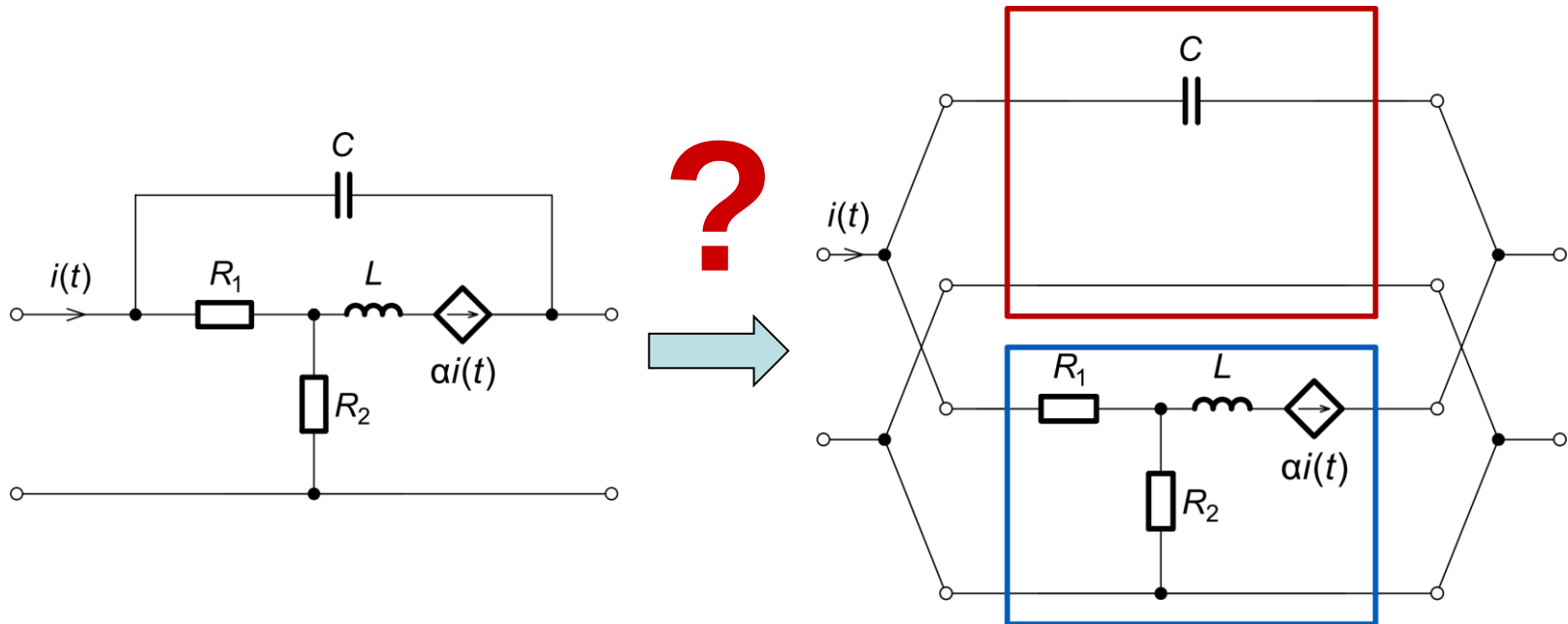
Complex impedance matrix we obtain after substituting $s = j\omega_0 = j$:

$$\mathbf{Z} = \begin{bmatrix} 3,8 + j0,4 & 1,8 + j0,4 \\ 4,2 - j0,4 & 2,2 + j0,6 \end{bmatrix}$$

Two-Port Networks

EXAMPLE 6

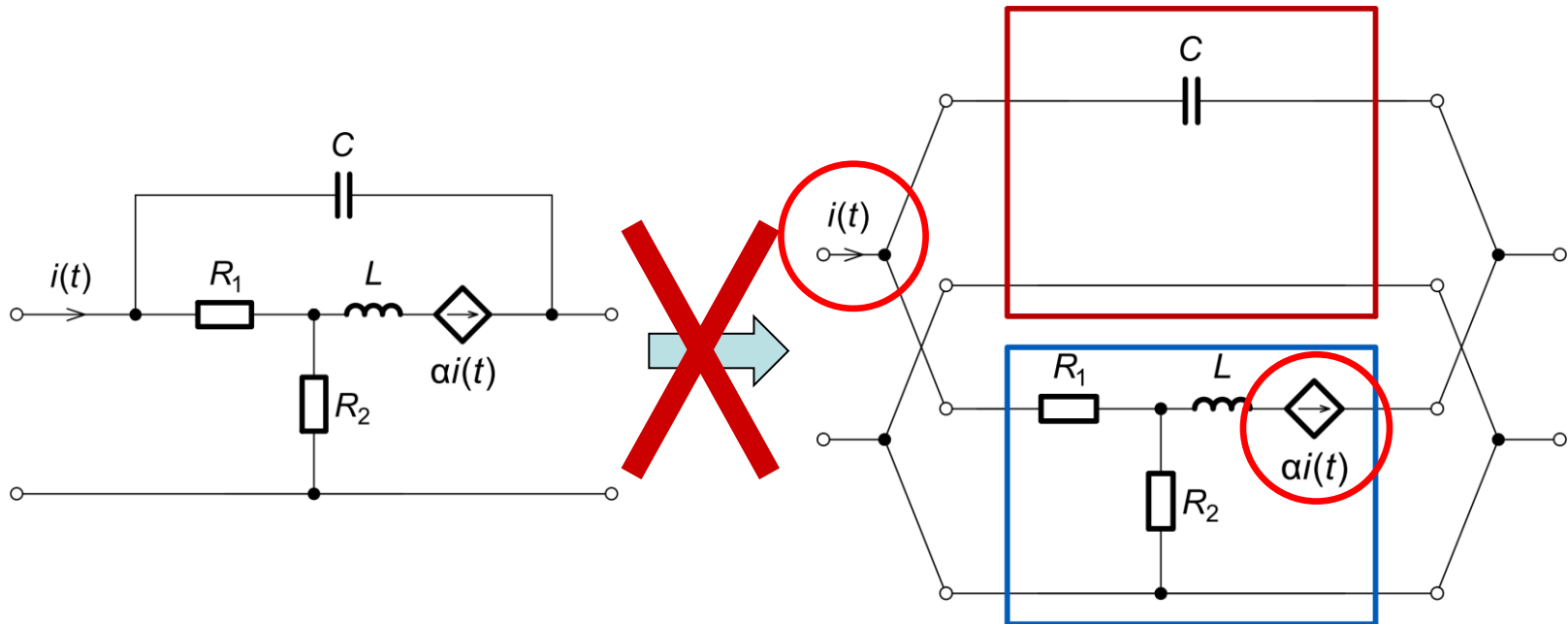
Question: Can the two-port be decomposed in a way as below?



Two-Port Networks

EXAMPLE 6

Question: Can the two-port be decomposed in such a way as below?

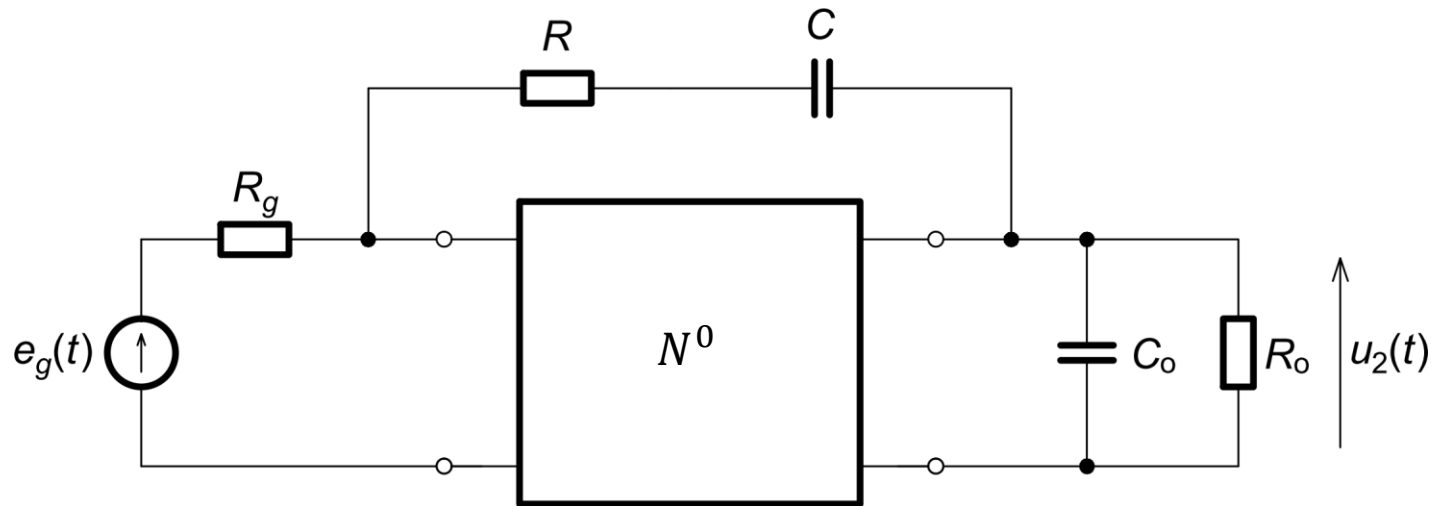


Answer: No, because in this configuration the steering value of the current-controlled voltage source is outside blue two-port, which makes the blue two-port's parameters dependent on external conditions. Hence, matrix representation for the blue two-port cannot be found.

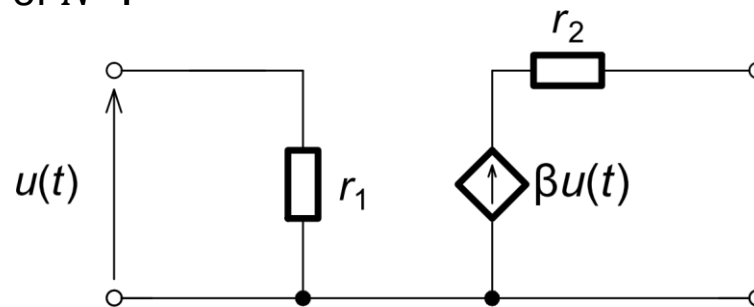
Two-Port Networks

EXAMPLE 7

Find RMS voltage amplification $K_{uRMS}(s) = \frac{U_2(s)}{E_g(s)}$ of a system shown below:



Equivalent system of N^0 :



$$R_g = 1\Omega, R_o = 1\Omega, C_o = \frac{1}{2}F, R = 2\Omega, C = \frac{1}{2}F, r_1 = 2\Omega, r_2 = \frac{1}{2}\Omega, \beta = 6$$

Two-Port Networks

EXAMPLE 7

Solution:

Admittance matrix of two-port network N^0 :

$$\mathbf{Y}^0(s) = \begin{bmatrix} \frac{1}{r_1} & 0 \\ -\frac{\beta}{r_2} & \frac{1}{r_2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ -12 & 2 \end{bmatrix}$$

After connecting RC branch admittance matrix of two-port network is:

$$\mathbf{Y}(s) = \mathbf{Y}^0(s) + \mathbf{Y}^{RC}(s) = \begin{bmatrix} \frac{1}{r_1} + \frac{1}{R + \frac{1}{sC}} & -\frac{1}{R + \frac{1}{sC}} \\ -\frac{\beta}{r_2} - \frac{1}{R + \frac{1}{sC}} & \frac{1}{r_2} + \frac{1}{R + \frac{1}{sC}} \end{bmatrix} = \begin{bmatrix} \frac{s + \frac{1}{2}}{s + 1} & -\frac{\frac{1}{2}s}{s + 1} \\ -\frac{\frac{25}{2}s + 12}{s + 1} & \frac{\frac{5}{2}s + 2}{s + 1} \end{bmatrix}$$

Two-Port Networks

EXAMPLE 7

RMS voltage amplification can be found from relationship:

$$K_{uRMS}(s) = \frac{y_{21}(s)Z_o(s)}{1 + y_{11}(s)Z_g(s) + y_{22}(s)Z_o(s) + Z_g(s)Z_o(s)\det Y(s)}$$

where:

$$Z_g(s) = R_g = 1, \quad Z_o(s) = \frac{1}{sC_o + \frac{1}{R_o}} = \frac{2}{s + 2}$$

After substitution we obtain:

$$K_{uRMS}(s) = \frac{\frac{25}{2}s + 12}{s^2 + \frac{3}{2}s + \frac{9}{2}}$$

System is BIBO stable, so after substituting $s = j\omega$ we get spectral characteristics:

$$K_{uRMS}(j\omega) = \frac{\frac{25}{2}j\omega + 12}{-\omega^2 + j\frac{3}{2}\omega + \frac{9}{2}}$$